Chapter 1

Discrete Mathematics

1.1 Objectives

Content adapted from [Piff(1991), Rosen(2007), Giunchiglia and Fumagalli(2017b)]. Please read first five chapters of [Piff(1991)] and first two chapters of [Rosen(2007)].

1. Defining Propositional Logic.
2. Defining Predicate Logic.
3. Defining Set, function, or relation model, and interpret the associated operations.
4. Operations between sets, functions, and relations.
5. Constructing proofs.
6. Logical reasoning to solve a strategic problem.
7. Model a real-life industry application applying formal methods.

1.2 Proposition Logic

Definition 1 (Proposition)**.** *A proposition is a statement that is either true or false.*

1.2.1 Examples of propositions

1. Blue is a color.
2. $1 + 1 = 2$.
3. Any two primes have a common factor.

1.2.2 Examples of sentences that are not propositions

1. What is the color of sky?
2. $1 + 1$.
3. This sentence is false.
4. $x^2 = x$.
5. There exists an x such that, $x^2 = x$.
6. For all integers x, $x^2 = x$.

Definition 2 (Propositional variables)**.** *A proposition or a boolean variable is a variable that is either true or false.*

Example 1. *The following symbolic sentence is read as x squared is equal to y.*

$$\underbrace{P(x, y)}_{\text{Predicates}} := \underline{x^2 = y}$$

. This is not a proposition.

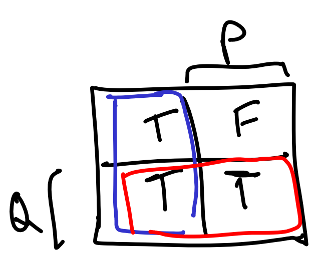
However, for a given x and y, the sentence becomes a proposition.

$$P(x = 1, y = 1) := 1^2 = 1$$

1.3 Propositional calculus aka Boolean Algebra

In propositional calculus, we have different names for Boolean algebra operators.

Boolean algebra	Propositional calculus
OR gate $x + y$	Disjunction: $P \vee Q$
AND gate $x \cdot y$	Conjunction: $P \wedge Q$
NOT gate $\bar{y}$	Negation: $\neg P$ P'



1.3.1 Implication $\Longrightarrow$, $\supset$, $\rightarrow$

$$(P \to Q) \equiv ((\underline{\neg P}) \vee Q)$$

P	Q	$P \to Q$
F	F	T
F	T	T
T	F	F ← M_2
T	T	T

$M_2 = \bar{P} + Q$

1.4 Propositional logic language

Definition 3. *Propositional alphabet*

① **Logic symbols** $\neg, \wedge, \vee, \supset, \equiv$

② **Propositional variables** $p \in \mathcal{P}$

③ **Separator symbols** *"(" and ")"*

Definition 4. *Well formed formulas*

Equivalance $\equiv \longleftrightarrow$

P	Q	$P \equiv Q$
F	F	T
F	T	F
T	F	F
T	T	T

$$(P \leftrightarrow Q) \equiv (\underbrace{\neg P \land \neg Q}_{m_0}) \lor (\underbrace{P \land Q}_{m_3})$$

- *every $p \in \mathcal{P}$ is an atomic formula*
- *if A and B are formulas then $\neg A$, $A \wedge B$, $A \vee B$, $A \supset B$ and $A \equiv B$ are formulas.*

Example 2. *Let's consider a propositional language where p means "Paola is happy", q means "Paola paints a picture", and r means "Renzo is happy". Formalize the following sentences:*

1. *"if Paola is happy and paints a picture" then Renzo isn't happy"* $(p \wedge q) \to \neg r$
2. *"if Paola is happy, then she paints a picture"* $p \to q$
3. *"Paola is happy" only if she paints a picture"* $q \leftrightarrow p$

The precision of formal languages avoid the ambiguities of natural languages.

Definition 5. *A Propositional interpretation is a function $I : P \to \{True, False\}$. (Evaluation of the formula with each variable set to true or false.)*

$(p \wedge q \to \neg r)$

p, q, r		
T F T	T	T
	T	F
	T	F
	T	F
	T	F

Tautology

Definition 6. *A formula A is*

Valid if for all interpretations I, $I \models A$ (All trues in TT)

(At least one T in the TT) *Satisfiable if there is an interpretations I s.t., $I \models A$*

Unsatisfiable if for no interpretations I, $I \models A$

(All F in the TT)

Propositional logic is an instrument for writing proofs. There are two connectives between two consecutive statements.

1.4.1 Equivalence $\Longleftrightarrow$

Example 3 ([Giunchiglia and Fumagalli(2017a)]). *Socrate says: "If I'm guilty, I must be punished; I must not be punished. Thus I'm not guilty." Is the argument logically correct?*

$$[(p \to q) \wedge (\neg q)] \to (\neg p)$$

p	q	$p \to q$	$(p\to q)\wedge\neg q$	$\neg p$
F	F	T	T	T
F	T	T	F	T
T	F	F	F	T
T	T	T	F	T

p	q	$p \to q$
F	F	T
F	T	T
T	F	F
T	T	T

p	q	$p \to q$
F	F	T
F	T	T
T	F	F
T	T	T

p	q	$\neg p$	$\neg q$
F	F	T	T
F	T	T	F
T	F	F	T
T	T	F	F

Example 4 ([Giunchiglia and Fumagalli(2017a)]). *Socrate says: "If I'm guilty, I must be punished; I must be punished. Thus I'm guilty." Is the argument logically correct?*

$$\boxed{((p \to q) \wedge q) \to p}$$

p	q	$(p \to q)$	$(p\to q)\wedge q$	p
F	F	T	F	T
F	T	T	T	F
T	F	F	F	T
T	T	T	T	T

not logically correct

Example 5. *As a reward for saving his daughter from pirates, the King has given you the opportunity to win a treasure hidden inside one of three trunks. The two trunks that do not hold the treasure are empty. To win, you must select the correct trunk. Trunks 1 and 2 are each inscribed with the message "This trunk is empty," and Trunk 3 is inscribed with the message "The treasure is in Trunk 2." The Queen, who never lies, tells you that only one of these inscriptions is true, while the other two are wrong. Which trunk should you select to win?*

p = "Treasure in T1"
q = "Treasure in T2"
r = "Treasure in T3"

1	2	3
"empty"	"empty"	Treasure in T2
$\neg p$	$\neg q$	q

$$\underbrace{[\neg p \wedge (\neg(\neg q)) \wedge (\neg q)]}_{\text{1 correct}} \vee \overbrace{[(\neg(\neg p)) \wedge \neg q \wedge \neg q]}^{\text{2 correct}} \vee \overbrace{[\neg(\neg p) \wedge \neg(\neg q) \wedge q]}^{\text{3 correct}}$$

$$[\neg p \wedge \underbrace{(q \wedge \neg q)}_{F}] \vee [p \wedge (\neg q) \wedge (\neg q)] \vee [p \wedge q \wedge q] = [F] \vee [p \wedge \neg q] \vee [p \wedge q]$$

Problem 1. *Suppose that in Example ~~7~~ 5 the inscriptions on Trunks 1, 2, and 3 are "The treasure is in Trunk 3," "The treasure is in Trunk 1," and "This trunk is empty." For each of these statements, determine whether the Queen who never lies could state this, and if so, which trunk the treasure is in.*
a) "All the inscriptions are false."
b) "Exactly one of the inscriptions is true."
c) "Exactly two of the inscriptions are true."
d) "All three inscriptions are true."

Problem 2 ([Giunchiglia and Fumagalli(2017a)]). *Kyle, Neal, and Grant find themselves trapped in a dark and cold dungeon (HOW they arrived there is another story). After a quick search the boys find three doors, the first one red, the second one blue, and the third one green. Behind one of the doors is a path to freedom. Behind the other two doors, however, is an evil fire-breathing dragon. Opening a door to the dragon means almost certain death.*

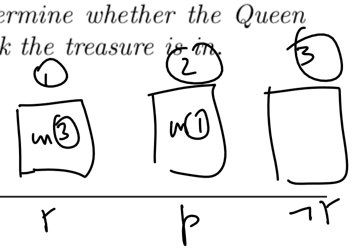
On each door there is an inscription:

freedom is behind this door	*freedom is not behind this door*	*freedom is not behind the blue door*

Given the fact that at LEAST ONE of the three statements on the three doors is true and at LEAST ONE of them is false, which door would lead the boys to safety?

Problem 1. *Suppose that in Example 5 the inscriptions on Trunks 1, 2, and 3 are "The treasure is in Trunk 3," "The treasure is in Trunk 1," and "This trunk is empty." For each of these statements, determine whether the Queen who never lies could state this, and if so, which trunk the treasure is in.*

a) *"All the inscriptions are false."*
b) *"Exactly one of the inscriptions is true."*
c) *"Exactly two of the inscriptions are true."*
d) *"All three inscriptions are true."*



p: Treasure in Trunk 1
q: " " " 2
r: " " " 3

"All inscriptions are false" $\equiv (\neg r) \wedge (\neg p) \wedge (\neg(\neg r))$

$= \neg r \wedge \neg p \wedge r$

$= F$

$\bar{x} \cdot x = 0$

"Exactly one inscription is true" $= (r \wedge (\neg p) \wedge (\neg(\neg r))) \vee (\neg r \wedge p \wedge (\neg(\neg r)))$

$\vee (\neg r \wedge \neg p \wedge \neg r)$

$p =$ "Treasure in T1"

$q =$ "Treasure in T2"

$r =$ "Treasure in T3"

Queens

$$= \underbrace{\left[\neg p \wedge (\neg(\neg q)) \wedge (\neg q)\right]}_{\text{1 correct}} \vee \overbrace{\left[(\neg(\neg p)) \wedge \neg q \wedge \neg q\right]}^{\text{2 correct}} \vee \overbrace{\left[\neg(\neg p) \wedge \neg(\neg q) \wedge q\right]}^{\text{3 correct}}$$

$$\neg p \qquad \neg q \qquad q$$

$$= \left[\neg p \wedge (\underbrace{q \wedge \neg q}_{F})\right] \vee \left[p \wedge (\neg q) \wedge (\neg q)\right] \vee \left[p \wedge q \wedge q\right]$$

$$= [F] \vee [p \wedge \neg q] \vee [p \wedge q] = p \wedge (\neg q \vee q) \qquad 0 + p\bar{q} + pq$$

$$= p \wedge T = p$$

Queen never lies $\Rightarrow p = T \Rightarrow$ Treasure is in Trunk 1

1.5 Predicates (variable depend propositions

Definition 7 (Predicates)**.** *A predicate is a statement $P(x_1, \ldots, x_n)$ involving n (not necessarily boolean) variables $x_1 \in \mathcal{X}_1$, $x_n \in \mathcal{X}_n$, and which evaluates to a proposition $P(x_1, \ldots, x_n) \in \{True, False\}$.*

Domain

Example 6. *Let $P(x)$ denote the statement "$x > 3$." What are the truth values of $P(4)$ and $P(2)$?*

$x = 4$ $\quad x = 2$

$P(4) = \text{True}, P(2) = \text{False}$

x is integer

$(\forall x. x > 3) = \text{False}$

1.5.1 Quantifiers

Domain $x = \{0, 1, 2, 3, 4\}$

Definition 8 (Universal quantifier $\forall$)**.** $\forall x : P(x) = P(0) \wedge P(1) \wedge P(2) \ldots$ $\forall x. P(x) \overset{?}{=}$ True or false

Definition 9 (Existential quantifier $\exists$)**.** there exists $(\exists x : P(x))$ $\quad (\exists x. x > 3) = \text{True}$

$\exists x. P(x) = P(0) \vee P(1) \vee P(2) \ldots$

Example 7. *Let $Q(x)$ be the statement "$x > 2$." What is the truth value of the quantification $\forall x Q(x)$, where the domain consists of all real numbers?*

Example 8. *Let $Q(x)$ denote the statement "$x = x + 1$." What is the truth value of the quantification $\exists x Q(x)$, where the domain consists of all real numbers?*

1.5.2 Rules of Inference for Propositional Logic

p: I am guilty

q: I must be punished

$p \to q$: If I am guilty then I must be punished

TABLE 1 Rules of Inference.

Rule of Inference	*Tautology*	*Name*
p $p \to q$ $\therefore q$	$[p \wedge (p \to q)] \to q$	Modus ponens
$\neg q$ $p \to q$ $\therefore \neg p$	$(\neg q \wedge (p \to q)) \to \neg p$	Modus tollens
$p \to q$ $q \to r$ $\therefore p \to r$	$((p \to q) \wedge (q \to r)) \to (p \to r)$	Hypothetical syllogism
$p \vee q$ $\neg p$ $\therefore q$	$((p \vee q) \wedge \neg p) \to q$	Disjunctive syllogism
p $\therefore p \vee q$	$p \to (p \vee q)$	Addition
$p \wedge q$ $\therefore p$	$(p \wedge q) \to p$	Simplification
p q $\therefore p \wedge q$	$((p) \wedge (q)) \to (p \wedge q)$	Conjunction
$p \vee q$ $\neg p \vee r$ $\therefore q \vee r$	$((p \vee q) \wedge (\neg p \vee r)) \to (q \vee r)$	Resolution

Given

$\therefore$ Therefore

① I will eat soup OR salad ($p \vee q$)

② I will not eat soup ($\neg p$)

$\to$ eat salad

$(p+q)(\bar{p}+r)$

$= p\bar{p} + q\bar{p} + pr + qr$

$= q\bar{p} + pr + qr$

$= \ldots$

① I will eat soup or salad

② I will either (not eat soup) or eat bread

I will eat salad or bread

Example 9. *Give a direct proof of the theorem "If n is an odd integer, then n^2 is odd."* $\forall n.$

$P(n)$: n is odd, Domain of n is integers

$\forall n. (P(n) \to P(n^2)) = T$

$P(n) = \exists k. \underbrace{(n = 2k+1)}_{Q(n,k)}$

Assume $P(n) = \text{True} \Rightarrow \exists k.\,(n = 2k+1) = \text{True} \Rightarrow$ Let k' be that integer

$$\begin{aligned}
P(n^2) &= \exists k.\,(n^2 = 2k+1) \\
&= \exists k.\,((2k'+1)^2 = 2k+1) \\
&= \exists k.\,(4k'^2 + 4k' + 1 = 2k+1) \\
&= \exists k.\,(2(2k'^2 + 2k') + 1 = 2k+1) \\
&= \text{True} \quad \Big|\ \text{for } k = 2k'^2 + 2k'
\end{aligned}$$

Hence proved $\forall n.\,(P(n) \rightarrow P(n^2))$

1.6 Sets

The concept of *set* is considered a primitive concept in math

A set is a collection of *elements* whose description must be unambiguous and unique: it must be possible to decide whether an element belongs to the set or not.

Prop "This statement" is false

member of set are element

Boolean variable {"True", "False"} {0,1} = {0,1,1} equality

1.6.1 Describing sets

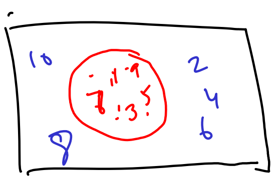
Listing $= \{0, 1, 2\}$, $B = \{0, \text{"a"}, \text{"hello"}, \{1\}, 3\}$,

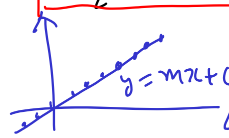
Predicate

Abstraction $N = \{0, 1, 2, \dots \infty\}$, $\boxed{O = \{n \in \mathbb{N} \mid P(n)\}}$

$\{n \in \mathbb{N} \mid \exists k. (n = 2k+1)\}$

Eulero-Venn Diagrams





$L_{(m,c)} = \{(x,y) \in \mathbb{R}^2 \mid y = mx + c\}$

set

1.6.2 Russell's paradox

Let R be the set of all sets that are not members of themselves (sometimes called "the Russell set"). If R is not a member of itself, then its definition entails that it is a member of itself; yet, if it is a member of itself, then it is not a member of itself, since it is the set of all sets that are not members of themselves. The resulting contradiction is Russell's paradox. In symbols:

$R = \{x \mid x \notin x\}$

$R \in R \Leftrightarrow R \notin R$

$$R = \{x \mid x \notin x\}. Then R \in R \iff R \notin R.$$

ZFC set theory

1.6.3 Relationships between sets

Definition 10 (Empty Set $\emptyset$)**.** $\emptyset = \{\}$ $= \emptyset$

Definition 11 (Singleton $\{a\}$)**.** $A = \{a\}$ *is singleton and* $\{a\} \neq a$.

is in   is not in

Definition 12 (Set membership $a \in A$)**.** $A = \{a, b, c\} \implies a \in A$ *and* $d \notin A$

Definition 13 (Set equality $A = B$)**.** $A = \{a, b, c\}$ $B = \{c, b, a, a\} \implies A = B$.

Definition 14 (Subset $A \subseteq B$)**.** $A = \{a, b, c\}$ $B = \{c, b, a, a\} \implies A \subseteq B$.
$A = \{a, b, c\}$ $B = \{c, b, a, a, d\} \implies A \subseteq B$.

$A \not\subset B$

Proper

Definition 15 (Subset $A \subset B$)**.** $A = \{a, b, c\}$ $B = \{c, b, a, a, d\} \implies A \subset B$.
$A = \{a, b, c\}$ $B = \{c, b, a, a\} \implies A \not\subset B$.

Definition 16 (Power set $P(A), 2^A$)**.** $A = \{a, b, c\}$, *then* $P(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a,$

Definition 17 (Union of two sets $A \cup B$)**.** $A = \{a, b, c\}$ $B = \{c, b, a, a, d\}$ $\implies A \cup B = \{a, b, c, d\}$.

Definition 18 (Intersection of two sets $A \cap B$)**.** $A = \{a, b, c\}$ $B = \{c, b, a, a, d\}$ $\implies A \cap B = \{a, b, c\}$.

Definition 19 (Set difference $A \setminus B$, $A - B$)**.** $A = \{a, b, c\}$ $B = \{c, b, a, a, d\}$
$\implies A \setminus B = A - B = \{\} = \emptyset$.
$B \setminus A = B - A = \{d\}$.

Definition 20 (Set complement $\bar{A}$, $C_U A$)**.** *Given a universal set* U*, complement is defined as* $\bar{A} = C_U A = U - A$*. Also,* $A \cup \bar{A} = U$*.*

Example 10. *Prove that* $U = (A \cap B) \cup \overline{(A \cap B)}$

Example 11. $A = \{a, e, i, o, \{u\}\}$ *and* $B = \{i, o, u\}$

1. $B \in A$
2. $(B - \{i, o\}) \in A$
3. $\{a\} \cup \{i\} \subset A$
4. $\{u\} \in A$
5. $\{\{u\}\} \in A$
6. $B - A = \emptyset$
7. $i \in A \cap B$
8. $\{i, o\} = A \cap B$

Example 12. *Prove De Morgan laws for sets* $\overline{A \cap B} = \bar{A} \cup \bar{B}$*, and* $\overline{A \cup B} = \bar{A} \cap \bar{B}$*.*

$$e \in A \cap B = e \in (A \cap B) = (e \in A) \wedge (e \in B) \tag{1.1}$$

$$e \in \overline{A \cap B} = \neg((e \in A) \wedge (e \in B)) = (\neg(e \in A) \vee \neg(e \in B)) = (e \notin A) \vee (e \in B) \tag{1.2}$$

$$e \in (\bar{A} \cup \bar{B}) = (e \in \bar{A}) \vee (e \in \bar{B}) = (e \notin A) \vee (e \notin B) \tag{1.3}$$

1.7 Relations

Definition 21 (Cartesian product of sets $A \times B$)**.** $A \times B = \{(a, b) : \forall a \in A, \forall b \in B\}$

Definition 22 (Relation $R \subseteq A \times B$)**.** *We write* aRy *to read as "a is R-related to b".*

Example 13. *given* $A = \{3, 5, 7\}$*,* $B = 2, 4, 6, 8, 10, 12$ *and* aRb *iff* a *is a divisor of* b*, then* $R = \{(3, 6), (3, 12), (5, 10)\}$

Definition 23 (Domain of R)**.**

Definition 24 (Image of R)**.**

Definition 25 (Inverse of R)**.**

1.8 Functions

Definition 26 (Function $f : A \to B$)**.** *Given two sets* A *and* B*, a function* f *from* A *to* B *is a relation that associates to each element* a *in* A *exactly one element* b *in* B*. Denoted with*
$f : A \to B$

1.8.1 Classes of Functions

Definition 27 (Surjective)**.**

Definition 28 (Injective)**.**

Definition 29 (Bijective)**.**

Definition 30 (Inverse)**.**

Definition 31 (Composition)**.**

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